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Batch: I13

Assignment 2

20 problem statements for a given dataset using NumPy and Pandas and Apply NumPy and Pandas methods to find the solution for the formulated p

Dataset:

Kaggle Text Classification Dataset

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[12] 47s

```
from google.colab import files
uploaded = files.upload()
```

Choose Files spam[2].csv

spam[2].csv(text/csv) - 503663 bytes, last modified: 22/4/2026 - 100% done

```
Saving spam[2].csv to spam[2].csv
```

[13] 0s

```
import pandas as pd
import numpy as np

df = pd.read_csv("spam[1].csv", encoding='latin-1')
df = df[['v1', 'v2']]
df.columns = ['label', 'text']
```

[14] 0s

```
#1. Total number of messages
df.shape[0]
```

```
5572
```

[15] 0s

```
#2. Number of spam vs ham messages
df['label'].value_counts()
```

	count
label	

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[15] 0s

```
#2. Number of spam vs ham messages
df['label'].value_counts()
```

	count
label	
ham	4825
spam	747

dtype: int64

[16] 0s

```
#3. Percentage of spam messages
(df['label'].value_counts(normalize=True) * 100)
```

	proportion
label	
ham	86.593683
spam	13.406317

dtype: float64

[17] 0s

```
#4. Check missing values
df.isnull().sum()
```

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[17]

```
#4. Check missing values
df.isnull().sum()
```

```
label    0
text     0
dtype: int64
```

[34]

```
#5. Count messages containing numbers
df['has_number'] = df['text'].str.contains(r'\d')
df['has_number'].sum()
```

```
np.int64(1456)
```

[35]

```
#6. Count messages containing URLs
df['has_url'] = df['text'].str.contains(r'http|www')
df['has_url'].sum()
```

```
np.int64(108)
```

[20]

```
#7. Longest message
df[df['length'] == df['length'].max()]
```

	label	text	length
1084	ham	For me the love should start with attraction.i...	910

[21]

```
#8. Shortest message
df[df['length'] == df['length'].min()]
```

	label	text	length
1924	ham	Ok	2
3049	ham	Ok	2
4496	ham	Ok	2
5357	ham	Ok	2

[22]

```
#9. Average message length
df['length'].mean()
```

```
np.float64(80.11880832735105)
```

[23]

```
#10. Compare spam vs ham length
df.groupby('label')['length'].mean()
```

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[23]

```python
#10. Compare spam vs ham length
df.groupby('label')['length'].mean()
```

	length
label	
ham	71.023627
spam	138.866131

dtype: float64

[36]

```python
#11. Percentage of messages above average length
avg_len = df['length'].mean()
(df['length'] > avg_len).mean() * 100
```

```
np.float64(38.85498923187365)
```

[25]

```python
#12. Average word count
df['word_count'].mean()
```

```
np.float64(15.494436468054559)
```

[26]

```python
#13. Most common words
from collections import Counter
words = " ".join(df['text']).split()
Counter(words).most_common(10)
```

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[26]

```python
#13. Most common words
from collections import Counter
words = " ".join(df['text']).split()
Counter(words).most_common(10)
```

```
[('to', 2134),
 ('you', 1622),
 ('I', 1466),
 ('a', 1327),
 ('the', 1197),
 ('and', 857),
 ('in', 798),
 ('is', 781),
 ('i', 742),
 ('u', 692)]
```

[27]

```python
#14. Most common words in spam
spam_words = " ".join(df[df['label']=='spam']['text']).split()
Counter(spam_words).most_common(10)
```

```
[('to', 604),
 ('a', 358),
 ('your', 187),
 ('call', 185),
 ('or', 185),
 ('the', 178),
 ('2', 169),
 ('for', 169),
 ('you', 164),
 ('is', 143)]
```

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[28]

```python
#15. Most common words in ham
ham_words = " ".join(df[df['label']=='ham']['text']).split()
Counter(ham_words).most_common(10)
```

```
[('to', 1530),
 ('you', 1458),
 ('I', 1436),
 ('the', 1019),
 ('a', 969),
 ('and', 738),
 ('i', 736),
 ('in', 734),
 ('u', 645),
 ('is', 638)]
```

[29]

```python
#16. Convert lengths to NumPy array
length_array = np.array(df['length'])
```

[30]

```python
#17. Mean and median length
np.mean(length_array), np.median(length_array)
```

```
(np.float64(80.11880832735105), np.float64(61.0))
```

[31]

```python
#18. Standard deviation of length
np.std(length_array)
```

```
np.float64(59.6854842151988)
```

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[32]

```python
#19. Messages longer than average
df[df['length'] > np.mean(length_array)]
```

	label	text	length	word_count
0	ham	Go until jurong point, crazy.. Available only ...	111	20
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	155	28
5	spam	FreeMsg Hey there darling it's been 3 week's n...	148	32
7	ham	As per your request 'Melle Melle (Oru Minnamin...	160	26
8	spam	WINNER!! As a valued network customer you have...	158	26
...	...	...	...	...
5557	ham	No. I meant the calculation is the same. That ...	273	48
5562	ham	Ok lor... Sony ericsson salesman... I ask shuh...	96	18
5566	spam	REMINDER FROM O2: To get 2.50 pounds free call...	147	28
5567	spam	This is the 2nd time we have tried 2 contact u...	161	30
5570	ham	The guy did some bitching but I acted like i'd...	125	26

2165 rows × 4 columns

[33]

```python
#20. Spam messages longer than average
df[(df['label']=='spam') & (df['length'] > np.mean(length_array))]
```

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RAM Disk

Share

[33]

```
#20. Spam messages longer than average
df[(df['label']=='spam') & (df['length'] > np.mean(length_array))]
```

	label	text	length	word_count
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	155	28
5	spam	FreeMsg Hey there darling it's been 3 week's n...	148	32
8	spam	WINNER!! As a valued network customer you have...	158	26
9	spam	Had your mobile 11 months or more? U R entitle...	154	29
11	spam	SIX chances to win CASH! From 100 to 20,000 po...	136	26
...	...	...	...	...
5537	spam	Want explicit SEX in 30 secs? Ring 02073162414...	90	16
5540	spam	ASKED 3MOBILE IF 0870 CHATLINES INCLU IN FREE ...	160	33
5547	spam	Had your contract mobile 11 Mnths? Latest Moto...	160	28
5566	spam	REMINDER FROM O2: To get 2.50 pounds free call...	147	28
5567	spam	This is the 2nd time we have tried 2 contact u...	161	30

698 rows × 4 columns

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